

Visualization 1 - Evolution of baby names over time

We want to understand how baby names evolved in France from 1900 to 2020: long-term popularity, sudden peaks, short trends, decline, comeback and recent trends.

10 brainstorming ideas:

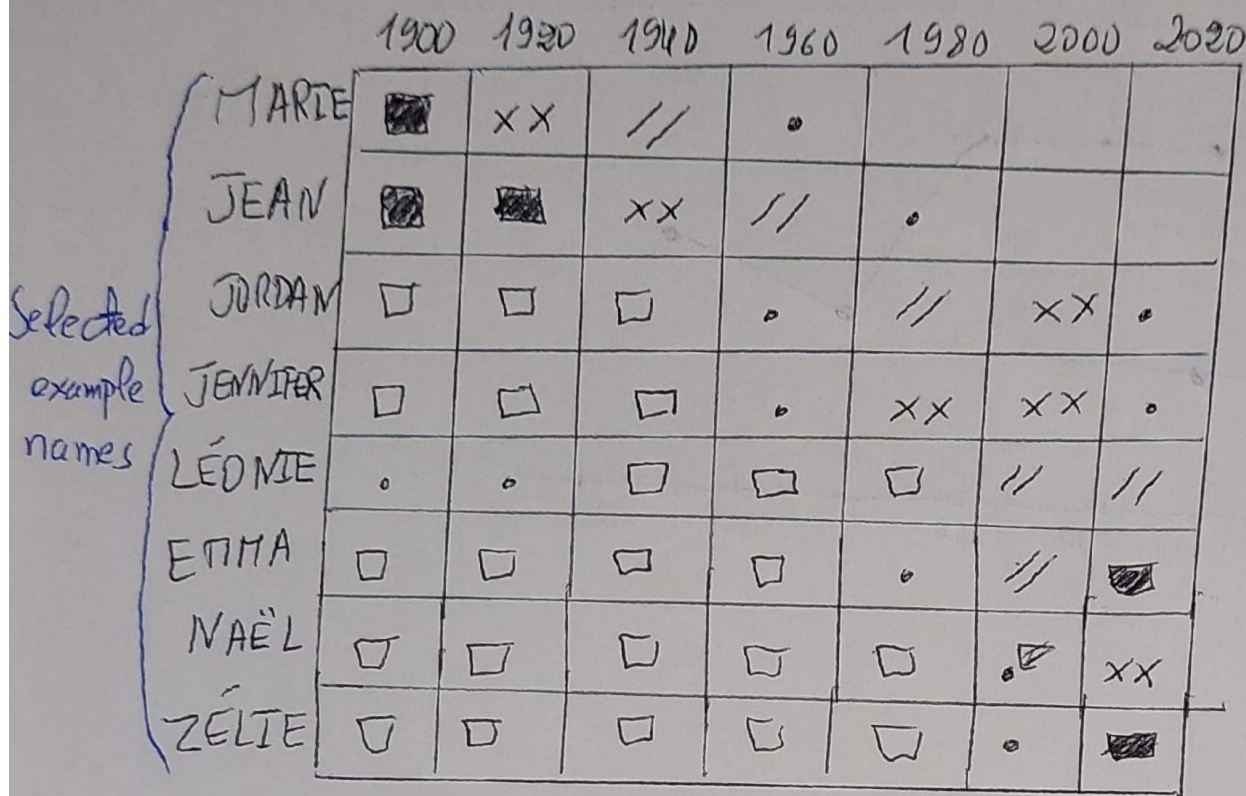
1. Line chart of selected names over time.
2. Heatmap name x year with color = normalized popularity
3. Streamgraph of dominant names over time
4. Bump chart showing rank changes over time
5. Small multiples: one mini-line chart per name
6. Peak timeline: each name placed at its peak year
7. Scatterplot: peak year vs peak intensity
8. Slope chart comparing two decades
9. Ridgeline chart showing popularity distributions by decade.
10. Interactive table + line chart for selected names

We selected three refined designs: heatmap, bump chart and annotated line chart

These ~~are~~ are design sketches based on a preliminary exploration of the dataset, not final computed visualizations.

Design A: Simplified heatmap of baby-name trajectories

Question: How do baby names evolve over time in France (1900-1920)?



Legend

□ = very low, • = low, // = medium, xx = high, ■ = very high

Pros: shows long-term trends, sudden peaks, come backs

Cons: too many names = clutter

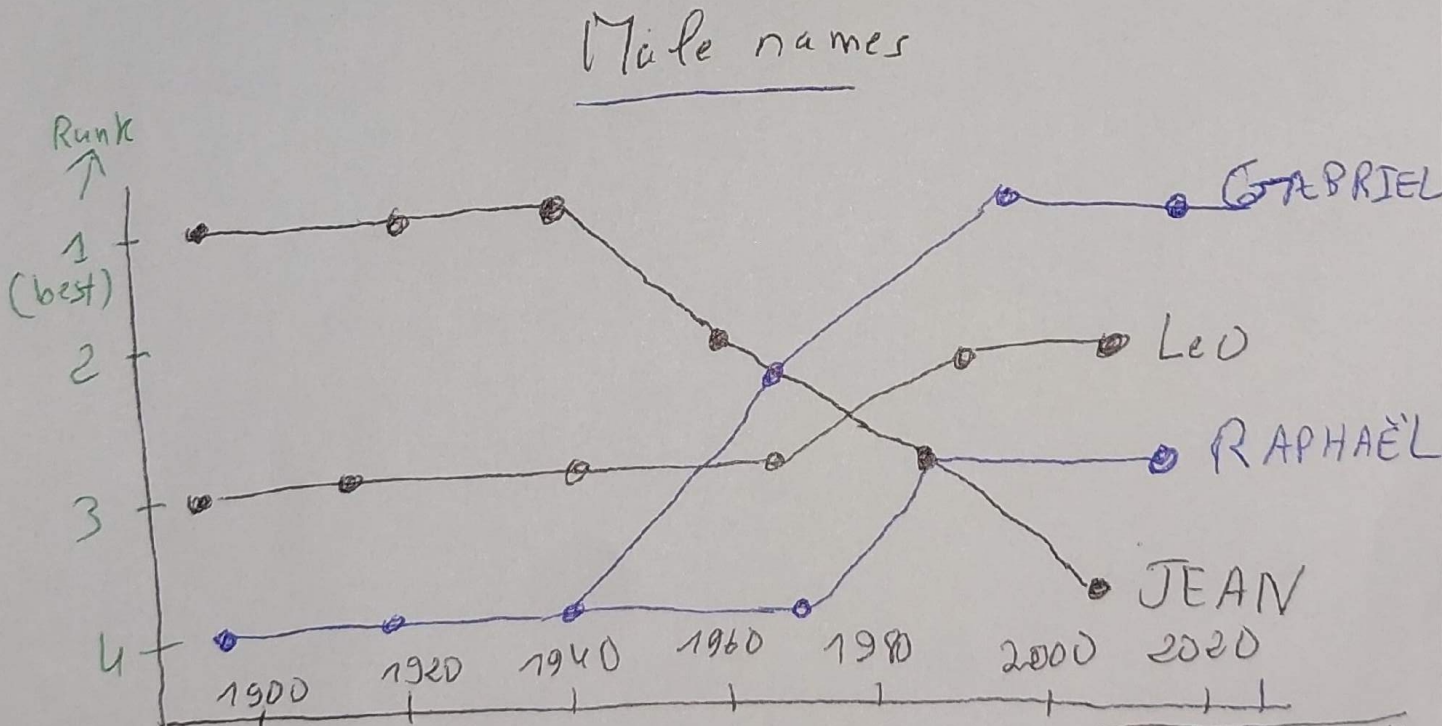
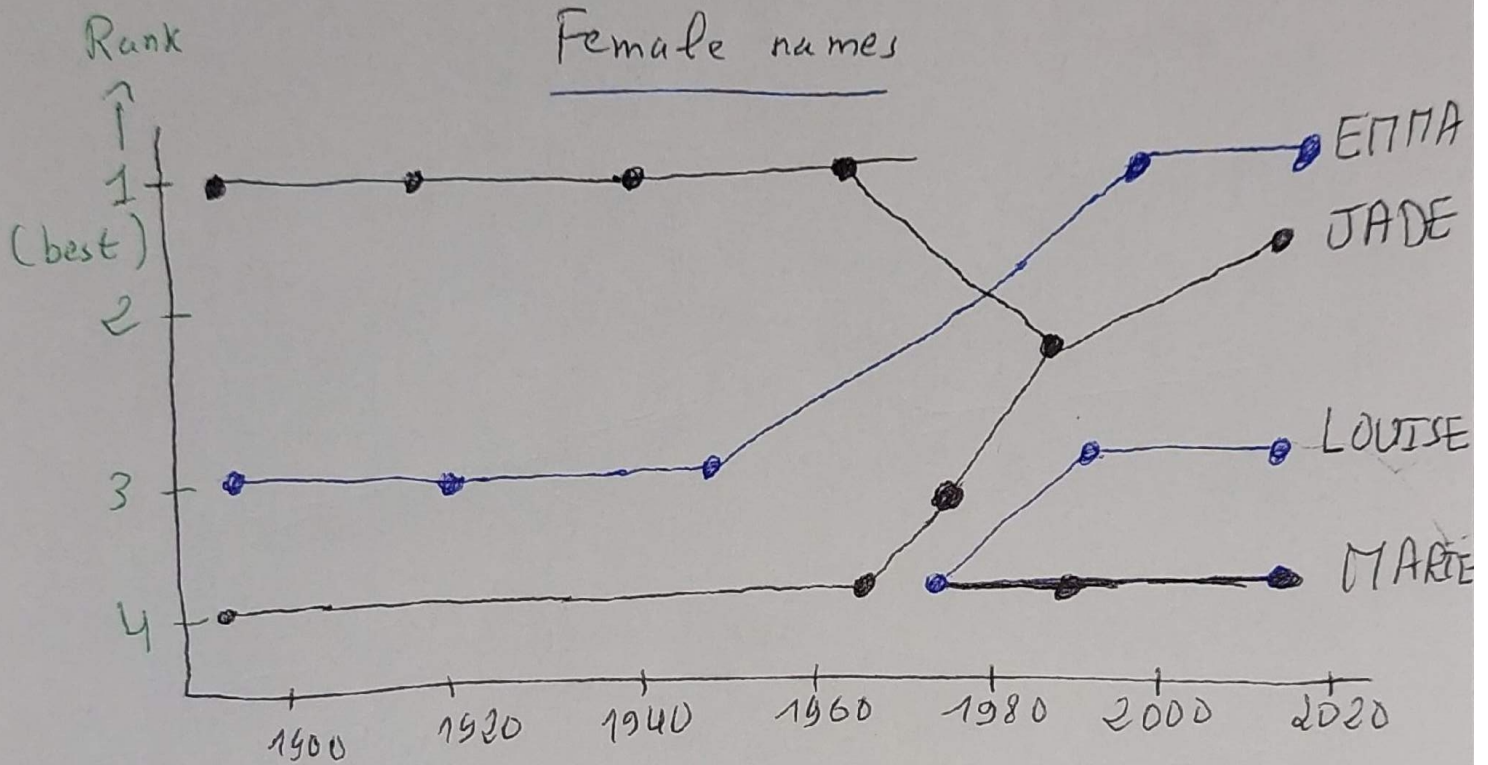
Possible interactions:

- search name
- filter by sex
- zoom years
- tooltip

This design gives a global view of how names evolve over time. It can show old popular names, sudden peaks, short lived trends and come backs. Its weakness it can become cluttered if too many names are shown. ~~We choose this as the final design, because it answers the question best~~

Design B: simplified bump chart of annual ranks

Question: Who dominates over time?



Pros: easy to see winners and rank changes

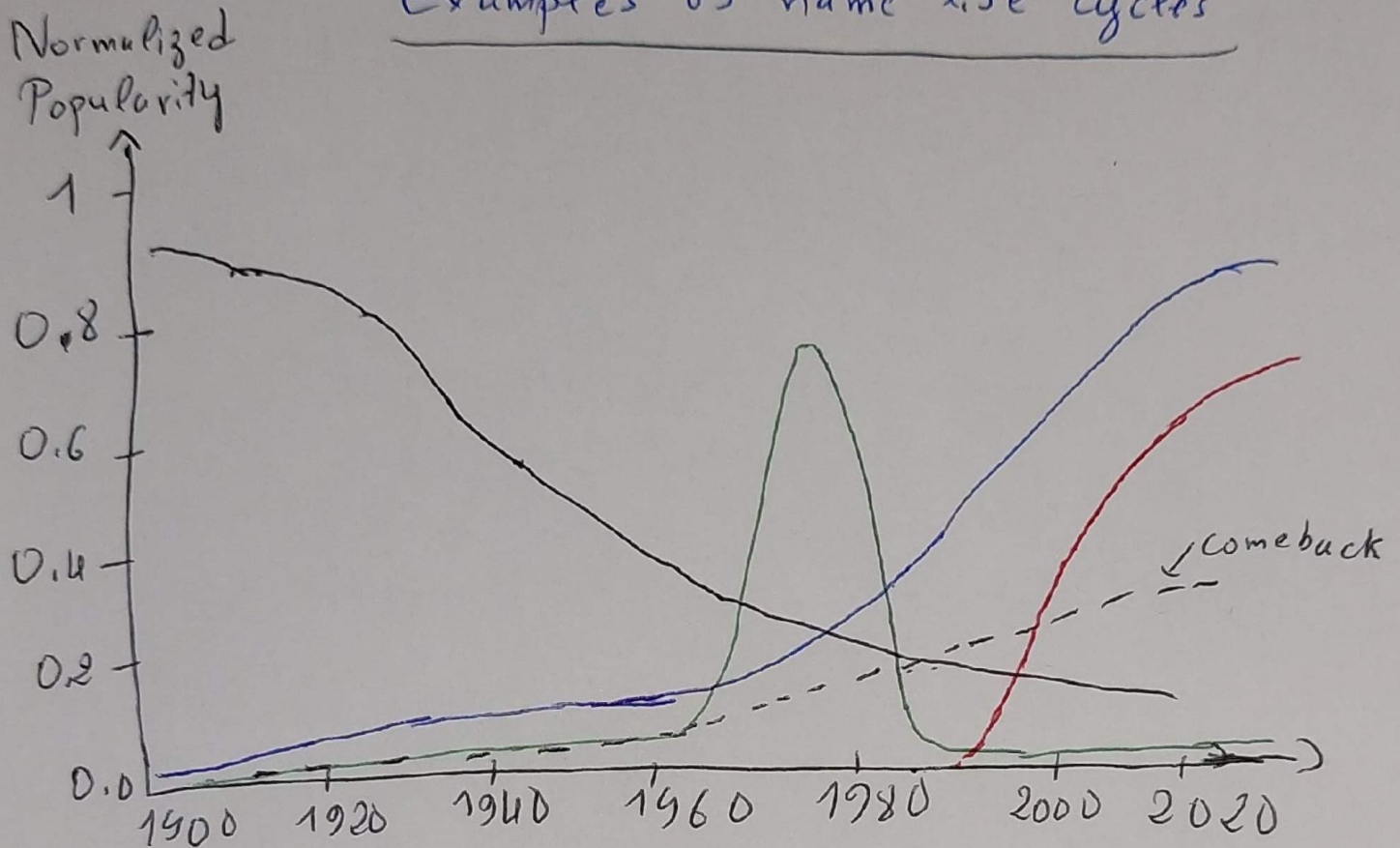
Cons: only shows Top N; not exactly popularity

Possible interactions

- hover highlight
- sex filter
- decade selection

Design C: Simplified annotated line chart

Examples of name life cycles



— MARIE
— ENNA
— JORDAN
— NAËL
--- LÉONINE

Pros: very intuitive, good for storytelling

Cons: works only for a few names

Possible interactions:

- select names
- tooltip
- separate M / F